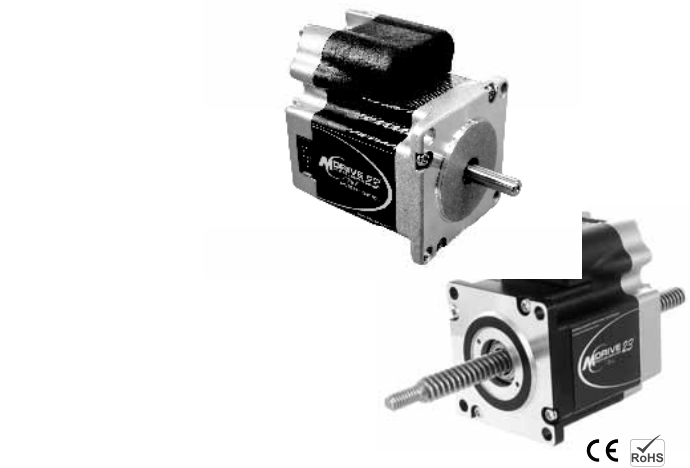


MDrive® 23
Motion Control



Notes and Warnings

Installation, configuration and maintenance must be carried out by qualified technicians only. You must have detailed information to be able to carry out this work. This information can be found in the user manuals.

- Unexpected dangers may be encountered when working with this product!
- Incorrect use may destroy this product and connected components!

The user manuals are not included. You can obtain them from the Internet at: <http://www.imshome.com/downloads/manuals.html>.

Required for Setup*

- PC running Microsoft® Windows XP Service Pack 2 or greater.
- IMS Terminal integrated program editor and terminal emulator (available online).
- +12 to +75 VDC (+12 to +60 VDC if using a quad length motor) unregulated linear or switching power supply.RS-422/485 communications interface (recommended: MD-CC400-001 or MD-CC402-001 Communication Converters). Or CANopen communications converter (recommended: MD-CC500-000).

Depending on your MDrive connectors configuration, you may also Power interface to 2-pin wire crimp connector (recommended: PD02-2300-FL3 prototype development cable).

- If using the 7-pin pluggable terminal we recommend 22 AWG shielded twisted pairs for logic wiring. Wire gauge for power connection varies with the distance from the MDrive and current. See MDrive product manual.
- I/O interface to 14-pin wire crimp connector. (Recommended: PD14-2334-FL3 prototype development cable)

* If you purchased your MDrive with a QuickStart Kit, you have received all of the connecting cables needed for initial functional setup and system testing.

Getting Started

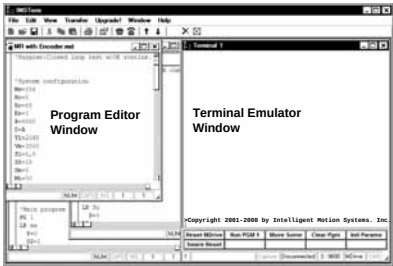
All documentation, software and resources are available online at: http://www.imshome.com/products/mdrive_motor_driver.html.

Connecting Power and I/O

Your MDrive is configured with power and I/O combined on a single connector. Please refer to the opposite side of this document for connecting details and available connectivity options including Prototype Development Cables and Mating Connector Kits.

Connecting Communications — RS-422/485

1. Connect RS-422/485 communications converter to MDrive and PC.
2. Install the communication converter drivers onto PC (available online).
3. Install and open IMS Terminal.
4. Apply power to MDrive.
5. Within IMS Terminal, click into the Terminal Window (shown below).
6. Key in CTRL+C. The MDrive sign-on message: “Copyright 2001-2008 by Intelligent Motion Systems, Inc.” should appear, verifying that communications is active.



Connecting Communications — CANopen

A “Getting Started” tutorial using the IMS CANopen Tester GUI with the MD-CC500-000 USB to CANopen dongle is located in the CANopen implementation manual, available online.

General Specifications

Electrical Specifications		
Input Voltage (+V) *	Single, Double and Triple Length	+12 to +75 VDC
	Quad Length	+12 to +60 VDC
Max Power Supply Current (Per MDrive 23)*	Single, Double and Triple Length	2 A
	Quad Length	3.5 A
Aux-Logic Input Voltage**		+12 to +24 VDC
Aux-Logic Input Current**		161 mA Max

*Actual Power Supply Current will depend on voltage and load.

**Used to power logic circuitry in the absence of +V.

Environmental Specifications

Operating Temperature (non-condensing)	Heat Sink	-40°C to +85°C
	Motor	-40°C to +100°C

I/O Specifications

General Purpose I/O - Number and Type	
I/O Points 1-4	4 I/O programmable as inputs (sinking or sourcing) or outputs (sinking)
I/O Points 1-4, 9-12 (Plus ² expanded feature)	8 I/O programmable as inputs or outputs (sinking or sourcing)
General Purpose I/O - Electrical	
Inputs	TTL up to +24 VDC
Sinking Outputs	Up to +24 VDC
Sourcing Outputs (Plus ² expanded feature)	+12 to +24 VDC
Output Sink Current	up to 600 mA (one channel)
Output Sink Current (Plus ² expanded feature)	up to 600 mA (one channel in each I/O bank)
Logic Threshold (Logic 0)	< 0.8 VDC
Logic Threshold (Logic 1)	> 2.2 VDC
Protection (Sinking)	Over Temp, Short Circuit
Protection (Sourcing)	Transient Over Voltage, Inductive Clamp
Analog Input	
Resolution	10 Bit
Range (Voltage Mode)	0 to +5 VDC, 0 to +10 VDC
Range (Current Mode)	4 to 20 mA, 0 to 20mA
Clock I/O	
Types	Step/Direction, Up/Down, Quadrature
Logic Threshold	+5V TTL Input, TTL Output (with 2 kΩ load to ground)
Trip Output/Capture Input	
Logic Threshold	+5V TTL Input, TTL Output (with 2 kΩ load to ground)

Communications Specifications

Protocol	RS-422/RS-485
BAUD Rate	4.8k, 9.6k, 19.2k, 38.4k, 115.2 kbps
CANopen Option	
Protocol	CAN 2.0B Active
Communications Profile	CIA DS-301
BAUD Rate	10, 20, 50, 125, 250, 500, 800 kBit/s, 1MBit/s (default)
Note: 800 kbps not supported by the MD-CC500-000 USB to CANopen dongle.	

Motion Specifications

Microstep Resolution - Open Loop	
Number of Resolutions	20

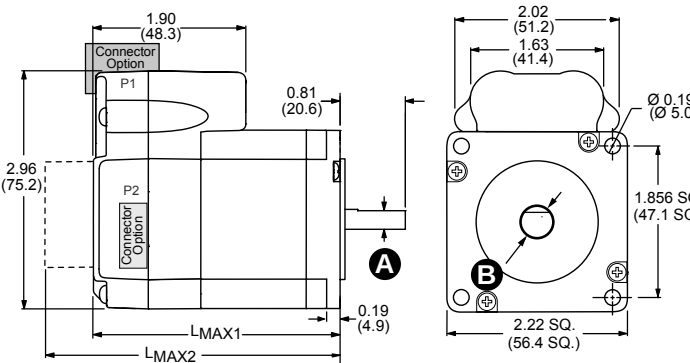
Available Microsteps Per Revolution								
200	400	800	1000	1600	2000	3200	5000	6400
12800	20000	25000	25600	40000	50000	51200	36000 ¹	21600 ²
1=0.01 deg/μstep 2=1 arc minute/μstep 3=0.001 mm/μstep								

Software Specifications

Program Storage Type/Size	Flash/6384 Bytes
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Mechanical Specifications

NOTE: For linear actuator products, see manual for screw specifications



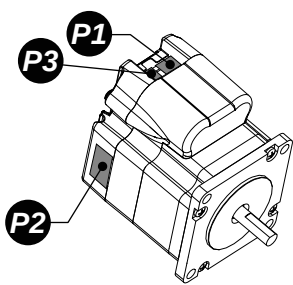
Motor Length	Dimensions in inches (mm)	
	LMAX (Single Shaft)	LMAX2 (Control Knob or External Encoder)
Single	2.65 (67.31)	3.36 (85.34)
Double	3.02 (76.71)	3.73 (94.74)
Triple	3.88 (98.55)	4.59 (116.59)
Quad	5.28 (134.15)	5.99 (152.19)

- A** Single, Double & Triple Length Motors:
0.230 ±0.004 (5.8 ±0.1)

Quad Length Motor:
0.2756 ±0.004 (7.0 ±0.1)
- B** Single, Double & Triple Length Motors:
Ø 0.2500 +0/-0.0005 (Ø 6.350 +0/-0.013)

Quad Length Motor:
Ø 0.315 +0/-0.0005 (Ø 8.0 +0/-0.013)

MDrive 23
Motion Control
Connectivity Options



P1 I/O & Power
Pluggable terminal or flying leads

Table with 2 columns: Wire Color, Function. Rows include White/Yellow (I/O1), White/Orange (I/O2), White/Violet (I/O3), White/Blue (I/O4), Green (Analog In), Black (Ground), Red (+V).

P1 I/O
14-pin wire crimp

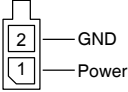
Table with 2 columns: Pin, Function. Rows include 7 CH A+, 8 CH A-, 9 CH B+, 10 CH B-, 13 IDX+, 14 IDX-.

Prototype Development Cable p/n: PD14-2334-FL3
Speed test and development with pre-wired mating connector.

Table with 4 columns: Pair, Wire Colors, Function, Encoder Function. Rows include 1 White Step Clock, 2 Green Capt/Trip, 3 Blue I/O11, 4 Black I/O12, 5 Yellow I/O9, 6 Brown I/O10, 7 Orange I/O3, 8 Black I/O4, 9 Red I/O1, 10 Black I/O2, 11 I/O Power, 12 I/O Ground.

Mating Connector Kit p/n: CK-09
Use to make your own cables, kit contains 5 mating connector shells with crimp pins. JST crimp tool recommended.
JST Parts Shell: PADD-14V-1-S Pins: SPH-001T-P0.5L

P3 Power
2-pin wire crimp



Prototype Development Cable p/n: PD02-2300-FL3
Function: Power Interface

Table with 2 columns: Wire Colors, Function. Rows include Black (Power Ground), Red (+V).

Mating Connector Kit p/n: CK-04
Use to make your own cables, kit contains 5 mating connector shells with crimp pins. Tyco crimp tool recommended.
Tyco Parts Shell: 794617-2 Pins: 794610-1

P2 Communications — CANopen version
DB-9 (male)

Communications Converter p/n: MD-CC500-000
Electrically isolated in-line USB to CANopen converter. USB "A" Type connector to DB-9 (Male). An interface cable must be constructed by the user.

Mating Cable Requirements
The following diagram illustrates the parts and connections for an interface cable connecting the MD-CC500-000 to the MDrive. Required Parts: (2) DB-9 (female), +7 to +30 VDC power supply, (1) 120Ω terminating resistor.

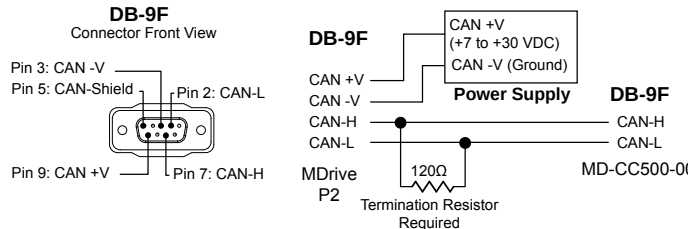
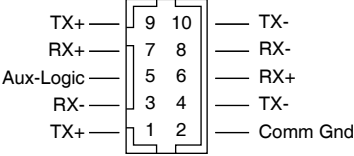
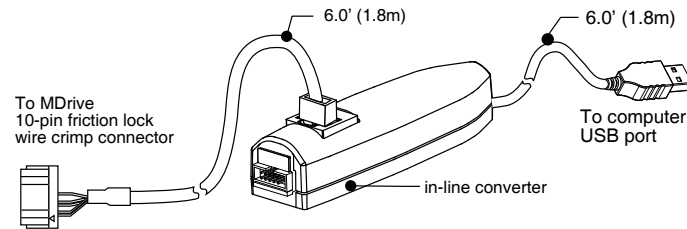


Table with 2 columns: Connector Style, Function. Rows include Pluggable Terminal (I/O and Power), Flying Leads (I/O and Power), 14-pin Wire Crimp (I/O), 10-pin Wire Crimp (Communications), 10-pin IDC (Communications), DB-9 (male) (Communications), 2-pin Wire Crimp (Power).

P2 Communications — RS-422/485
10-pin wire crimp



Communications Converter p/n: MD-CC402-001
Electrically isolated in-line USB to RS-422/485 converter pre-wired with mating connector to conveniently program and set configuration parameters.

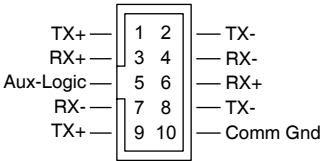


Prototype Development Cable p/n: PD10-1434-FL3
Speed test and development with pre-wired mating connector. Recommended for multi-drop systems, can be used in conjunction with the MD-CC402-001.

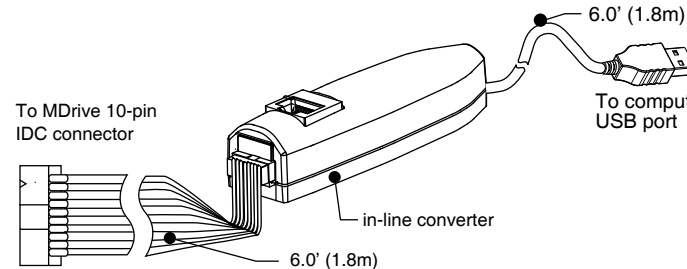
Table with 2 columns: Wire Colors, Function. Rows include White/Red Stripe (Aux-Logic), White/Blue Stripe (TX+), Blue/White Stripe (TX-), White/Orange Stripe (RX+), Orange/White Stripe (RX-), Green/White Stripe (GND).

Mating Connector Kit p/n: CK-02
Use to make your own cables, kit contains 5 mating connector shells with crimp pins. Hirose crimp tool recommended.
Hirose Parts Shell: DF11-10DS-2C Pins: DF11-2428SC

P2 Communications — RS-422/485
10-pin IDC



Communications Converter p/n: MD-CC400-001
Electrically isolated in-line USB to RS-422/485 converter pre-wired with mating connector to conveniently program and set configuration parameters.



Mating Connector Kit p/n: CK-01
Use to make your own cables, kit contains 5 mating connector shells for making interface cables.
IDC Parts Shell: SAMTEC TCSD-05-01-N Ribbon Cable: AMP 1-57051-9